

ADIT RAHMAN

613-263-2672 | arahmana@uwaterloo.ca | linkedin.com/in/adit-rahman | github.com/adit-rah

EDUCATION

University of Waterloo

Bachelor of Computer Science, Co-op | GPA: 3.81/4.00

Waterloo, Ontario

Sept. 2024 - Apr. 2028

- **Relevant Coursework:** Algorithms & Data Structures, Compilers, Databases, Numerical Computation

PROFESSIONAL EXPERIENCE

Faire

Software Engineer Intern

Sept. 2026 – Dec. 2026

San Francisco, California

- Incoming on the Security Engineering team, working on backend infrastructure in Kotlin and Kubernetes

Shopify

Software Engineer Intern

Jan. 2026 – Apr. 2026

Ottawa, Ontario

- Designed and shipped a production job system using Node.js, BullMQ, Redis, and Kubernetes worker pods to process **122K+ daily tasks** with custom Prometheus metrics and OpenTelemetry instrumentation
- Migrated production tables from Vitess to YugabyteDB for Shopify's ads domain, processing **11.6M records** with zero-downtime across a system serving **18M+ ads/month**
- Unblocked a cross-team Airflow DAG involved in ad delivery within hours by diagnosing a **130K-row** data inconsistency, tracing the root cause to composite primary key duplication
- Built the Product Network app's first staging environment with Docker, Kubernetes, and Buildkite CI/CD, enabling pre-production validation and catching breaking changes before release

Shopify

Software Engineer Intern

Sept. 2025 – Dec. 2025

Ottawa, Ontario

- Rebuilt the merchant eligibility pipeline with parallel validation and a decoupled cache layer for safe migration off an external dependency, sustaining **p50 32ms** latency across **10.2M** lookups at a **97.4%** cache hit rate
- Decoupled the ads-publisher service from the merchant model by introducing a request abstraction and updating GraphQL resolvers, unlocking onboarding for **230K** merchants previously blocked by the legacy integration
- Built centralized shop disablement logic handling 581 merchant lifecycle transitions, consolidating previously fragmented merchant lifecycle management
- Optimized the merchant earnings page with client-side data fetching, reducing initial load time by **8x**

PROJECTS

Nanofable 🌀 | *PyTorch, Python, JavaScript, Hugging Face*

- Trained **8** custom transformer LMs from scratch (**16** runs, **8B** tokens) on **2x NVIDIA Tesla T4** GPUs, implementing the architecture with fp16 mixed precision on tensor cores to fit within per-device VRAM limits
- Implemented **1.58-bit ternary quantization-aware training** with straight-through estimators against fp16 baselines, locating the byte floor of coherent English
- Built a JS inference engine with a **packed ternary format**, reducing model size **6x** while preserving bit-identical outputs against the PyTorch reference, plus an evaluation harness scoring **4,200+** completions

Lock-In Vision Pipeline 🌀 | *Python, PyTorch, OpenCV, Streamlit, SQLite*

- Built a **real-time** focus detection system using a fine-tuned ResNet trained on the State Farm distracted-driver dataset (**22K+ images**) to classify focused vs. distracted behavior from webcam input, achieving **0.86 F1**
- Implemented an **on-device inference pipeline** with PyTorch and OpenCV using temporal prediction smoothing for stable real-time feedback, with Streamlit and SQLite supporting session tracking and persistence

Chess Engine 🌀 | *C++20, SDL2, WebAssembly*

- Developed a cross-platform chess engine in C++20 with SDL2 graphics rendering and a WebAssembly browser build, using object-oriented programming and Observer pattern for modular multi-display architecture
- Implemented 5 AI difficulty levels with 4-ply alpha-beta pruning, piece-square evaluation, and quiescence search, cutting evaluated states by **95%** and move computation from 4s to **~200ms**

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript/TypeScript, Go, Ruby, SQL, Bash

Infrastructure: Linux, Docker, Kubernetes, Git, Buildkite CI/CD, Redis, Airflow, MySQL, PostgreSQL, Prometheus, Grafana, OpenTelemetry

ML & Web: PyTorch, NumPy, OpenCV, Hugging Face, React, Node.js, GraphQL